

Polychlorinated Biphenyl-77 Induces Adipocyte Differentiation and Proinflammatory Adipokines and Promotes Obesity and Atherosclerosis

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Background Obesity, an inflammatory condition linked to cardiovascular disease, is associated with expansion of adipose tissue. Highly prevalent coplanar polychlorinated biphenyls (PCBs) such as 3,3',4,4'-tetrachlorobiphenyl (PCB-77) accumulate in adipose tissue because of their lipophilicity and increase with obesity. However, the effects of PCBs on adipocytes, obesity, and obesity-associated cardiovascular disease are unknown. Objectives In this study we examined in vitro and in vivo effects of PCB-77 on adipocyte differentiation, proinflammatory adipokines, adipocyte morphology, body weight, serum lipids, and atherosclerosis. Methods PCB-77 or 2,2',4,4,5,5'-hexachlorobiphenyl (PCB-153) was incubated with 3T3-L1 adipocytes either during differentiation or in mature adipocytes. Concentration-dependent effects of PCB-77 were contrasted with those of 2,3,7,8-tetrachlorodibenzo-p-dioxin (TCDD). For in vivo studies, we treated C57BL/6 wild-type (WT) or aryl hydrocarbon receptor (AhR)-/- mice with vehicle or PCB-77 (49 mg/kg, by intraperitoneal injection) and examined body weight gain. In separate studies, we injected ApoE-/- mice with vehicle or PCB-77 over a 6-week period and examined body weight, adipocyte size, serum lipids, and atherosclerosis. Results Low concentrations of PCB-77 or TCDD increased adipocyte differentiation, glycerol-3-phosphate dehydrogenase activity, and expression of peroxisome proliferator-activated receptor γ , whereas higher concentrations inhibited adipocyte differentiation. Effects of PCB-77 were abolished by the AhR antagonist α -naphthoflavone. PCB-77 promoted the expression and release of various proinflammatory cytokines from 3T3-L1 adipocytes. Administration of PCB-77 increased body weight gain in WT but not AhR-/- mice. ApoE-/- mice injected with PCB-77 exhibited greater body weight, adipocyte hypertrophy, serum dyslipidemia, and augmented atherosclerosis. Conclusions Our findings suggest that PCB-77 may contribute to the development of obesity and obesity-associated atherosclerosis.